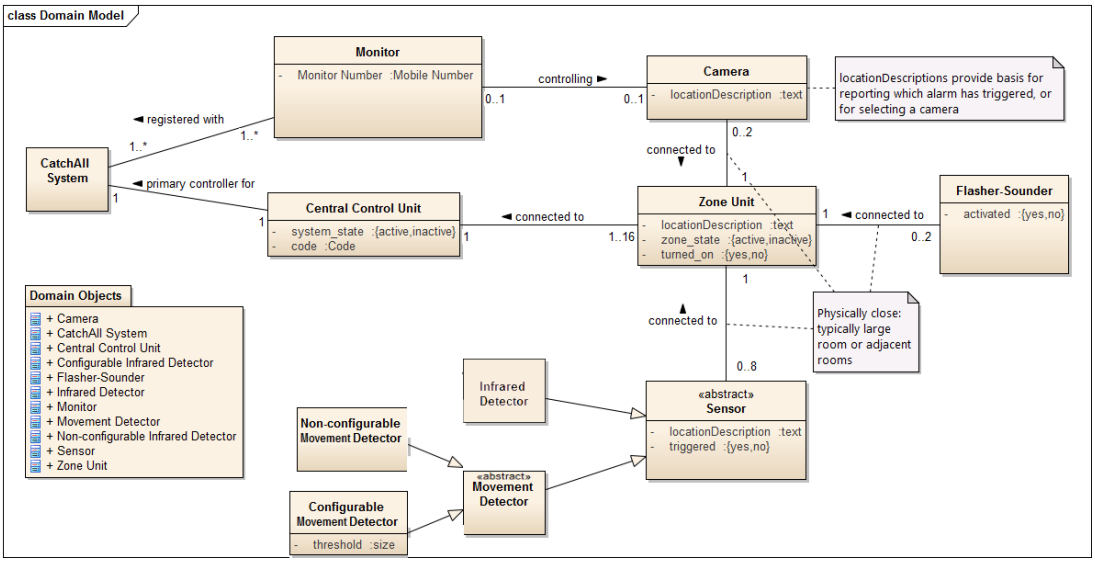
SWEN30006 Software Modelling and Design
Workshop 11: Getting a Better GRASP on Design - Partial Solution

School of Computing and Information Systems
University of Melbourne
Semester 1, 2025

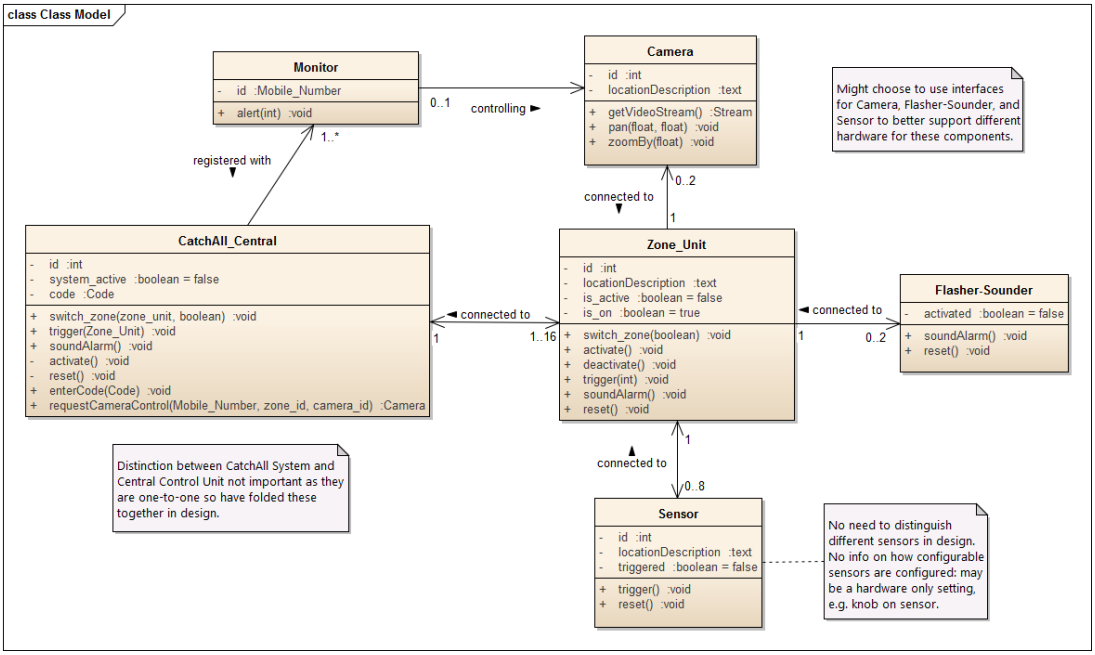
Note: your solutions should be hand-drawn. Your models may differ subject to assumptions you have made.

Part 1 Part A

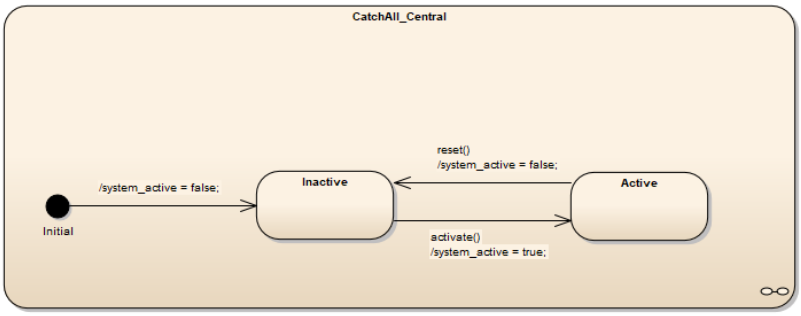
Q1.1 Exercise 1

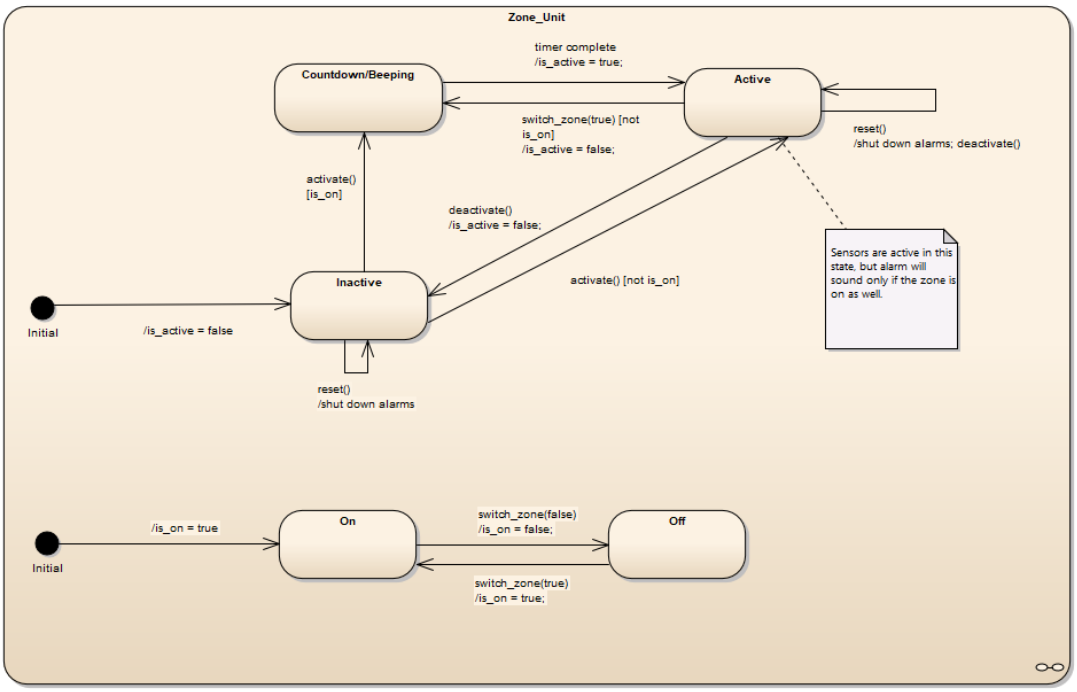


Q1.2 Exercise 2



Q1.3 Exercise 3

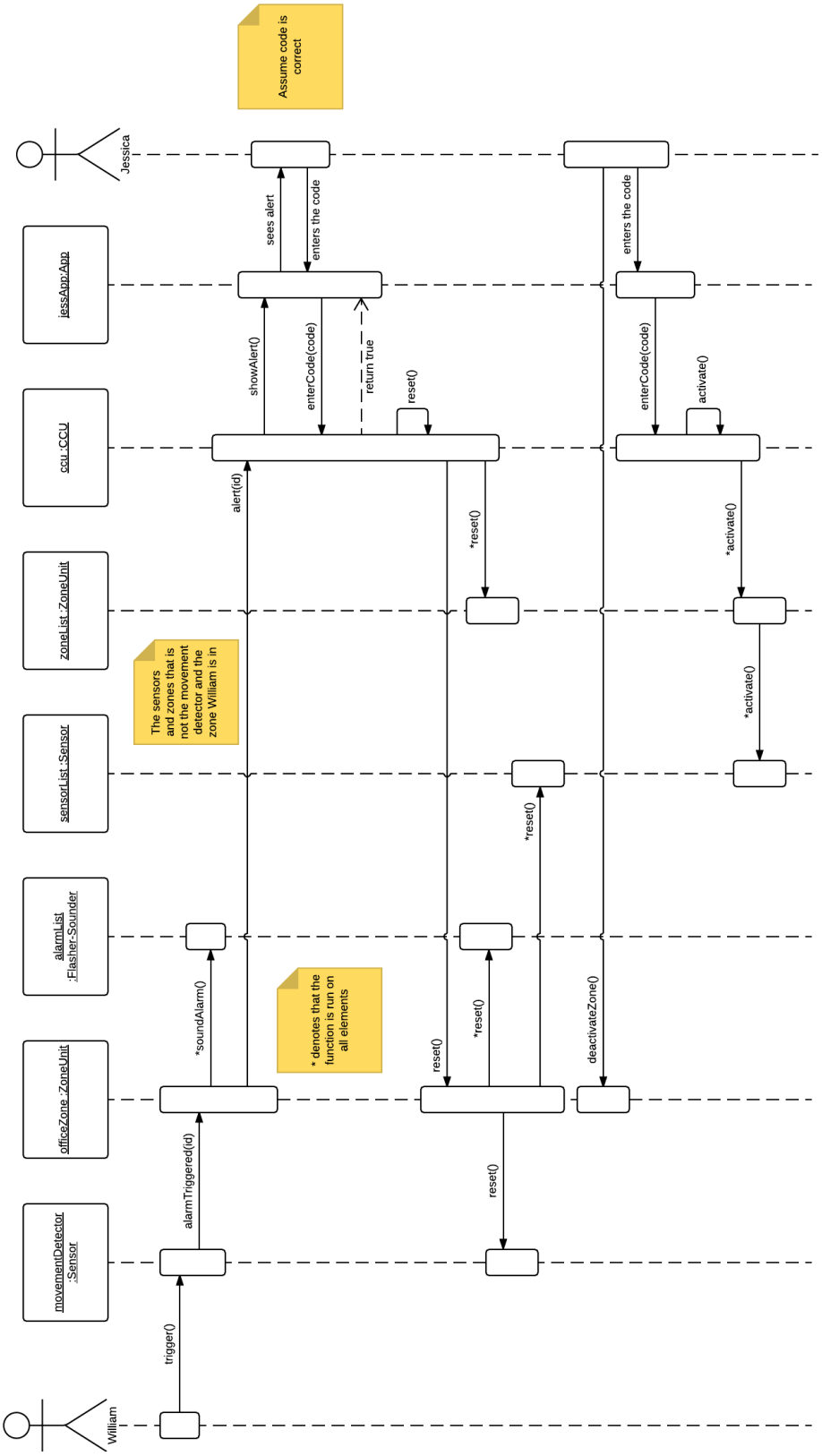




Part 2 Part B

Q2.1 Exercise 4

This is really 4 sequence diagrams in one: (1) trigger (2) alert (3) reset (4) deactivatezone/activate.


William
Jessica
movementDetector :Sensor
officeZone :ZoneUnit
alarmList :Flasher-Sounder
sensorList :Sensor
zoneList :ZoneUnit
ccu :CCU
jessApp:App
trigger()
alarmTriggered(id)
*soundAlarm()
alert(id)
showAlert()
sees alert
enters the code
enterCode(code)
return true
reset()
*reset()
reset()
*reset()
*reset()
reset()
deactivateZone()
enters the code
enterCode(code)
activate()
*activate()
*activate()
Assume code is correct
The sensors and zones that is not the movement detector and the zone William is in
* denotes that the function is run on all elements